MATHS ASSIGNMENT- II

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1. A machine that inserts letters into envelopes goes haywire and inserts randomly into envelopes. What is the probability that in a group of 100 letters

a) no letter is put into the correct envelope.
b) Exactly 1 letter is put into the correct envelope.
c) Exactly 98 letters are put into correct envelope.
d) Exactly 99 letters are put into correct envelope.
e) All letters are put into the correct envelopes.

a) The probability of no letter being put in the correct envelop is $\frac{d_{100}}{100!}$

b) When exactly one letter is put correctly, the no. of derangements for the remaining 99 letters is d_{99}. This can happen in ${}^{100}C_1$ ways. So the required probability is $\frac{{}^{100}C_1 \times d_{99}}{100!}$

c) $\frac{{}^{100}C_{98} \times d_2}{100!}$; when exactly 98 letters are put into the right envelop, the no. of derangements for the remaining two is d_2.

d) When exactly 99 letters are put into the correct envelop, it is not possible to misplace the remaining 1 letter. This is also put into the right one. Thus the probability is zero.

e) When all the letters are put into the correct envelop, this can be happen only out of 100! cases. Thus the required probability is $\frac{1}{100!}$

2. Find the rook polynomial for the standard 3 x 3 chess board.

=> Here, $r_0 = 1$

$r_1 = 9$

$r_2 = 18$

$r_3 = 6$

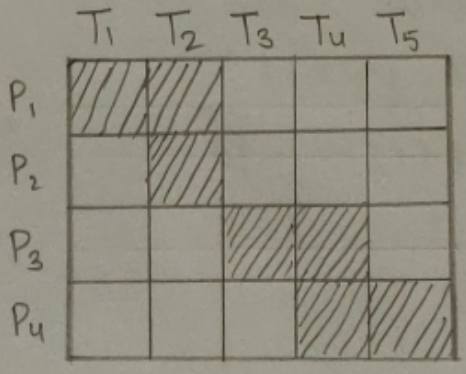
$r_4 = 0$

1	2	3
4	5	6
7	8	9

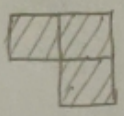
$\Rightarrow 1 + 9x + 18x^2 + 6x^3$

3. Four persons P_1, P_2, P_3, P_4 who arrive late for dinner party find that only one chair at each of the five tables T_1, T_2, T_3, T_4 and T_5 is vacant. P_1 will not sit at T_1 or T_2, P_2 will not sit at T_2, P_3 will not sit at T_3 or T_4, and P_4 will not sit at T_4 or T_5. Find the number of ways they can occupy the vacant chairs.

=>

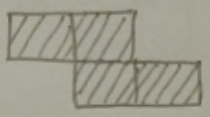


The shaded squares in the first row indicate the tables T_1 & T_2 are forbidden for P_1. Find the rook polynomial for the shaded squares. Here there exists 2 disjoint sub-boards

C_1: 

$r_0 = 1,\ r_1 = 3,\ r_2 = 1,\ r_3 = 0.$

$\Rightarrow 1 + 3x + x^2$

C_2: 

$r_0 = 1,\ r_1 = 4,\ r_2 = 3$

$\Rightarrow 1 + 4x + 3x^2$

$r(C, x) = r(C_1, x) \times r(C_2 x)$

$= (1 + 3x + x^2)(1 + 4x + 3x^2)$

$= 1 + 7x + 16x^2 + 13x^3 + 3x^4$

$\downarrow\ r_0 \quad \downarrow\ r_1 \quad \downarrow\ r_2 \quad \downarrow\ r_3 \quad \downarrow\ r_4.$

$S_0 = n! = 5!$

$S_k = (n-k)!\, r_k$

$S_1 = (5-1)!\, r_1$

$= 4! \times 7$

$S_2 = (5-2)!\, r_2$

$= 3! \times 16$

$S_3 = (5-3)!\, r_3$

$= 2! \times 13$

$S_4 = (5-4)!\, r_4$

$= 3$

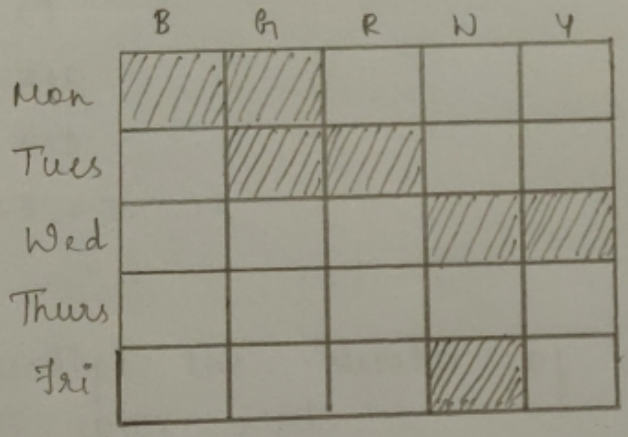
Consequently the no. of ways in which four persons can occupy the chairs is

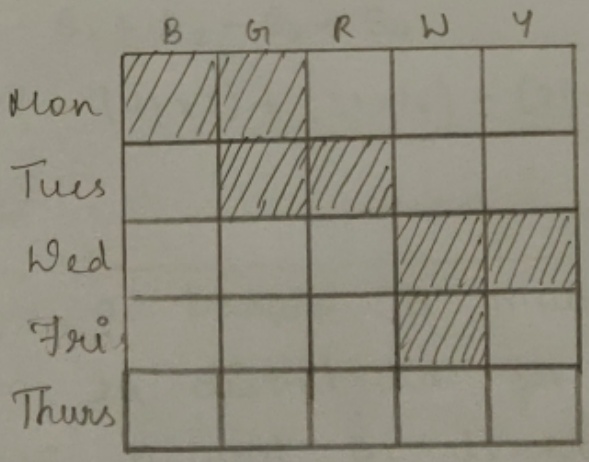
$5! - 4!(7) + 3!(16) - 2!(13) + 3$

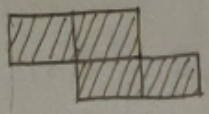
$=$

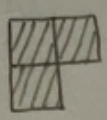
4. An executive attending a week long business meeting has 5 suits of different colors: Blue, Green, Red, White and Yellow. On Mondays he doesn't wear Blue or Green, on Tuesdays he doesn't wear Read or Green, on Wednesdays he doesn't wear White or Yellow and on Fridays he doesn't wear White. In how many ways can he dress without repeating a color during the meeting?

=>





C_1 : 

C_2 : 

$r_0 = 1, r_1 = 4, r_2 = 3$

$1 + 4x + 3x^2$

$r_0 = 1, r_1 = 3, r_2 = 1$

$1 + 3x + x^2$

$$r(C,x) = r(C_1,x) \times r(C_2,x)$$

$$= (1+4x+3x^2)(1+3x+x^2)$$

$$= 1 + 7x + 16x^2 + 13x^3 + 3x^4$$

$r_0 \quad r_1 \quad r_2 \quad r_3 \quad r_4$

$n! = 5! = S_0$

$S_k = (n-k)!\, r_k$

$= (5-k)!\, r_k$

$S_1 = 4! \times 7$

$S_2 = 3! \times 16$

$S_3 = 2! \times 13$

$S_4 = 1 \times 3 = 3$

$S_5 = 0$

Consequently the number of ways he can dress with 5 color suits

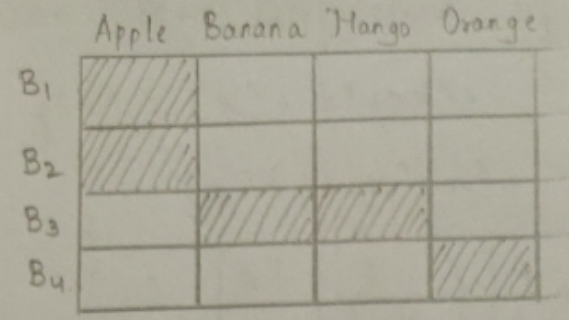
$\overline{N} = S_0 - S_1 + S_2 - S_3 + S_4$

$= 5! - (4! \times 7) + (3! \times 16) - (2! \times 13) + 3$

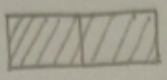
$= \underline{\underline{25}}$

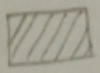
5. An apple, a banana, a mango and an orange are to be distributed for four boys B_1, B_2, B_3, B_4. The boys B_1 and B_2 do not wish to have apple, the boy B_3 does not want banana or mango and B_4 refuses orange. In how many ways the distribution can be made so that no boy is displeased?

$\Rightarrow$



C_1: 

C_2: 

C_3: 

$r_0 = 1, r_1 = 2, r_2 = 0$

$r(C_1, x) = 1 + 2x$

$r_0 = 1, r_1 = 2$

$r(C_2, x) = 1 + 2x$

$r_0 = 1, r_1 = 1$

$r(C_3, x) = 1 + x.$

$\therefore r(C, x) = r(C_1 x) \times r(C_2 x) \times r(C_3 x)$

$= (1+2x)^2 (1+x)$

$= 1 + 5x + 8x^2 + 4x^3$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$

$r_0 \quad r_1 \quad r_2 \quad r_3$

$n! = 4!$

$S_0 = 4!$

$S_k = (n-k)! r_k$

$S_1 = (4-1)! r_1$

$= 4! \times 5$

$S_2 = (4-2)! r_2$

$= 2! \times 8$

$S_3 = (4-3)! r_3$

$= 4$

$\overline{N} = S_0 - S_1 + S_2 - S_3 + S_4$

$= \underline{\underline{6}}$

6. Determine the coefficient of x^{50} in $f(x) = (x^7 + x^8 + x^9 + \dots)^6$

$\Rightarrow$ Let $f(x) = (x^7 + x^8 + x^9 + \dots)^6$

$= [x^7(1 + x^7 + x^8 + \dots)]^6$

$= \left[x^7 \left(\frac{1}{1-x}\right)\right]^6$

$= \frac{x^{42}}{(1-x)^6}$

$= x^{42}(1-x)^{-6}$

$$f(x) = x^{42} \sum_{r=0}^{\infty} {}^{-6}C_r (-x)^r$$

$$= x^{42} \sum_{r=0}^{\infty} (-1)^r \, {}^{6+r-1}C_r \, (-1)^r x^r$$

$$= \sum_{r=0}^{\infty} {}^{5+r}C_r \, x^{r+42}$$

For $r = 8$, we have.

$$5 + {}^{8}C_8 \, x^{8+42} = {}^{13}C_8 \, x^{50}$$

$\therefore$ The coeff of x^{50} is ${}^{13}C_8$

7. Determine the coefficient of x^{20} in $f(x) = (x^2 + x^3 + x^4 + x^5 + x^6)^5$.

$\Rightarrow$ $f(x) = (x^2 + x^3 + x^4 + x^5 + x^6)^5$

$= [x^2(1 + x + x^2 + x^3 + x^4)]^5$

$$= \left[x^2\left(\frac{1-x^5}{1-x}\right)\right]^5$$

$$= \frac{x^{10}(1-x^5)^5}{(1-x)^5}$$

$$= x^{10} \sum_{r=0}^{5} {}^5C_r (-x^5)^r \sum_{k=0}^{\infty} {}^{-5}C_k (-x)^k$$

$$= x^{10} \sum_{r=0}^{5} {}^5C_r (-1)^r x^{5r} \sum_{k=0}^{\infty} (-1)^k {}^{5+k-1}C_k (-1)^k x^k$$

$$= x^{10} \sum_{r=0}^{5} {}^5C_r (-1)^r x^{5r} \sum_{k=0}^{\infty} {}^{4+k}C_k x^k$$

case ① = $r=2$ and $k=0$.

$$x^{10}\ {}^5C_2 (-1)^2 x^{10} = {}^5C_2 x^{20}$$

Case ② = $r=0$ and $k=10$

$$x^{10}\ {}^{4+10}C_{10} x^{10} = {}^{14}C_{10} x^{20}$$

case ③ = $r=1$ and $k=5$ we have

$$x^{10}\ {}^5C_2 (-1) x^5\ {}^{4+5}C_5 x^5$$

$${}^{-5}C_1 \times {}^9C_5 \times x^{20} = {}^5C_2 + {}^{14}C_{10} - ({}^5C_1 \times {}^9C_5)$$

$$=$$

8. In how many ways can 12 marbles be distributed among three students A, B, C so that A gets atleast 4, B and C gets atleast 2 so that C gets not more than 5 marbles.

$A = x^4 + x^5 + x^6 + \ldots = x^4[1 + x + x^2 + \ldots]$

$= \frac{x^2}{1-x}.$

$B = x^2 + x^3 + x^4 + \ldots = \frac{x^2}{1-x}.$

$C = x^2 + x^3 + x^4 + x^5 = x^2(1 + x + x^2 + x^3)$

$= x^2\left[\frac{1-x^4}{1-x}\right]$

$\therefore f(x) = \frac{x^2}{1-x} \times \frac{x^2}{1-x} \times x^2\left[\frac{1-x^4}{1-x}\right]$

$= \frac{x^8(1-x^4)}{(1-x)^3}$

$f(x) = (x^8 - x^{12})(1-x)^{-3}$

$= x^8 \sum_{r=0}^{\infty} {}^{-3}C_r (-x)^r - x^{12} \sum_{k=0}^{\infty} {}^{-3}C_k (x)^k$

$f(x) = x^8 \sum_{r=0}^{\infty} (-1)^r \; {}^{3+r-1}C_r (-1)^r x^r - x^{12} \sum_{k=0}^{\infty} (-1)^k \; {}^{3+k-1}C_r (-1)^k x^k$

$f(x) = x^8 \sum_{r=0}^{\infty} {}^{2+r}C_1 \, x^r - x^{12} \sum_{k=0}^{\infty} {}^{2+k}C_k \, x^k$

For $r=4$ and $k=0$, we have

$x^8 \; {}^{2+4}C_4 x^4 - x^{12} = ({}^6C_4 - 1)x^{12}.$

$\therefore$ $({}^6C_4 - 1)$ ways to distribute the 12 marbles to 3 students of desired type.

9. In how many ways can two dozen identical robots be assigned to four assembly line with

i) at least three robots assigned to each line?

ii) at least three, but not more than nine robots assigned to each line?

$\Rightarrow$

i) Given that $f(x) = (x^3 + x^4 + x^5 + \dots)^4$

$$= [x^3(1 + x + x^2 + \dots)]^4$$

$$= \left(x^3\left[\frac{1}{1-x}\right]\right)^4 = \frac{x^{12}}{(1-x)^4}$$

$$f(x) = x^{12}(1-x)^{-4}$$

$$= x^{12} \sum_{r=0}^{\infty} {}^{-4}C_r (-x)^r$$

$$= x^{12} \sum_{r=0}^{\infty} (-1)^r \; {}^{4+r-1}C_r (-1)^r x^r$$

$$= x^{12} \sum_{r=0}^{\infty} {}^{3+r}C_r \, x^r$$

for $r = 12$, $x^{12} \; {}^{3+12}C_{12} \, x^{12} = {}^{15}C_{12} \, x^{24}$.

ii) $f(x) = (x^3 + x^4 + x^5 + \dots + x^9)^4$

$$= [x^3(1 + x + x^2 + \dots + x^6)]^4$$

$$= \left[x^3\left(\frac{1-x^7}{1-x}\right)\right]^4$$

$$= \frac{x^{12}(1-x^7)^4}{(1-x)^4}$$

$$f(x) = x^{12}(1-x^7)^4(1-x)^{-4}$$

$$= x^{12} \sum_{r=0}^{\infty} {}^4C_r (-x^7)^r \sum_{k=0}^{\infty} {}^{-4}C_k (-x)^k$$

$$= x^{12} \sum_{r=0}^{\infty} (-1)^r {}^4C_r x^{7r} \sum_{k=0}^{\infty} {}^{3+k}C_k x^k$$

<u>case 1</u> = For $r=0$ and $k=12$

$$x^{12} \times 1 \times {}^{3+12}C_{12} \times x^{12} = {}^{15}C_{12} x^{24}$$

<u>case 2</u> = For $r=1$ and $k=5$

$$x^{12}(-1)\,{}^4C_1 x^7 \times {}^{3+5}C_5 x^5$$

$$-{}^4C_1 \times {}^8C_5 x^{24}$$

$\therefore$ ${}^{15}C_{12} - ({}^4C_1 \times {}^8C_5)$

10. In how many ways 3000 identical envelopes be divided in packages of 25, among four student groups so that each group gets at least 150 but not more than 1000 of the envelopes.

$\Rightarrow$ The number of packets = $\frac{3000}{25} = 120$.

Each group gets atleast $\frac{150}{25}$ envelopes = 6 packets

but not more than $\frac{1000}{25}$ envelopes = 40 packets

∴ The generating function for the given problem is

$$f(x) = (x^6 + x^7 + \dots + x^{40})^4$$

$$= [x^6(1 + x + x^2 + \dots + x^{34})]^4$$

$$= \left[x^6\left(\frac{1 - x^{35}}{1 - x}\right)\right]^4$$

$$= x^{24}\left(\frac{1 - x^{35}}{1 - x}\right)^4 \Rightarrow x^{24}(1 - x^{35})^4 (1 - x)^{-4}$$

$$f(x) = x^{24} \sum_{r=0}^{4} {}^4C_r (-x^{35})^r \sum_{k=0}^{\infty} {}^{-4}C_k (-x)^k$$

$$= x^{24} \sum_{r=0}^{4} {}^4C_r (-1)^r x^{35r} \sum_{k=0}^{\infty} {}^{3+k}C_k x^k$$

<u>case 1</u> = for $r = 0$ and $k = 96$, we have

$$x^{24} \times 1 \times {}^{3+96}C_{96} x^{96} = {}^{99}C_{96} x^{120}$$

<u>case 2</u> = for $r = 1$ and $k = 61$, we have

$$x^{24} \, {}^4C_1 (-1) x^{35} \times {}^{3+61}C_{61} \times x^{61} = {}^{-4}C_2 \, {}^{64}C_{61} x^{120}$$

<u>case 3</u> = for $r = 2$ and $k = 26$ we have

$$x^{24} \, {}^4C_2 (-1)^2 x^{70} \times {}^{3+26}C_{26} x^{26} = {}^4C_2 \times {}^{29}C_{26} x^{120}.$$

The coefficient of x^{120} is

$${}^{99}C_{96} + ({}^{-4}C_2 \times {}^{64}C_{61}) + ({}^4C_2 \times {}^{29}C_{26})$$

11. A ship carries 48 flags, 12 each of colors red, white, blue and black. Twelve of these flags are placed on a vertical pole in order to communicate a signal to other ships

i) How many ways of these signals use an even number of blue flags and odd number of black flags?

ii) How many of the signals have at least 3 white flags or no white flags at all?

$\Rightarrow$ $f_1(x) = x^0 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots = \frac{1}{2}[e^x + e^{-x}]$

$f_2(x) = \frac{x}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots = \frac{1}{2}[e^x - e^{-x}]$

$f_3(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = e^x = f_4(x).$

$f(x) = f_1(x) \times f_2(x) \times f_3(x) \times f_4(x)$

$= \frac{1}{4}(e^{2x} - e^{-2x})(e^{2x})$

$= \frac{1}{4}(e^{4x} - 1)$

$= \frac{1}{4} e^{4x} - \frac{1}{4}$

$= \frac{1}{4} \sum_{r=0}^{\infty} \frac{(4x)^r}{r!} - \frac{1}{4}.$

The coefficient of $\frac{x^{12}}{12!} = \frac{1}{4} \times 4^{12} = \underline{\underline{4^{11}}}$

ii) In this case we have

$$f(x) = \left(x^0 + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots\right)\left(x^0 + x + \frac{x^2}{2!} + \dots\right)^3$$

$$= \left[1 + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots + x + \frac{x^2}{2!} - x - \frac{x^2}{2!}\right](e^x)^3$$

$$= \left[e^x - x - \frac{x^2}{2!}\right]e^{3x}$$

$$f(x) = e^{4x} - x.e^{3x} - \frac{x^2}{2!}e^{3x}$$

$$f(x) = \sum_{r=0}^{\infty} \frac{(4x)^r}{r!} - x\sum_{k=0}^{\infty} \frac{(3x)^k}{k!} - \frac{x^2}{2}\sum_{l=0}^{\infty} \frac{(3x)^l}{l!}$$

For $r = 12$, $k = 11$ and $l = 10$ we have.

$$= \left[4^{12} \times \frac{x^{12}}{12!}\right] - x\left[3^{11} \times \frac{x^{11}}{11!}\right] - \frac{x^2}{2}\left[\frac{3^{10} \times x^{10}}{10!}\right]$$

$$= \left[4^{12} \times \frac{x^{12}}{12!}\right] - \left[3^{11} \times \frac{x^{12}}{12!}\right] - \frac{1}{2}\left[3^{10} \times \frac{x^{12}}{10!}\right]$$

$$= \left[4^{12} - (3^{11} \times 12) - (3^{10} \times 6 \times 11)\right]\frac{x^{12}}{12!}$$

The cost of $\frac{x^{12}}{12!} = 4^{12} - (3^{11} \times 12) - (3^{10} \times 66)$

$=$

12. Using the exponential generating function, find the number of ways in which 4 of the letters in the word ENGINE be arranged.

⇒ We have E → 2, N → 2, G = I = 1.

In an arrangement of these letters the letter E can appear 0, 1 or 2 times. The exponential generating function corresponding to letter E is

$$f_1(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!}$$

Similarly the Exponential generating to letter N is

$$f_2(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!}$$

Since the G can appear 0 or 1 time in an arrangement then the corresponding exponential generating function is $f_3(x) = 1 + \frac{x}{1!}$

Also the letter I can appear 0 or 1 time

Hence $f_4(x) = 1 + \frac{x}{1!}$

$$f(x) = f_1(x)\, f_2(x)\, f_3(x)\, f_4(x)$$

$$= \left(1 + x + \frac{x^2}{2!}\right)(1+x)^2$$

$$= \left(1 + 2x + 2x^3 + x^3 + \frac{x^4}{4!}\right)\left(1 + 2x + x^2 + 0x^3 + 0x^4\right)$$

$$= x^4\left[(1\times 0) + (2\times 0) + (1\times 2) + (1/4 \times 1)\right]$$

$$= x^4\left[2 + 2 + 1/4\right] = x^4 \times 4!\,(17/4).$$

∴ The coefficient of $\frac{x^4}{4!} = 4! \times \frac{17}{4} = \underline{\underline{102}}$